

Citation addendum to “*The Sturmian–Mahler Edge of the Accelerated Collatz Realizer Problem*”

(June 2026)

Elias De Jesús

Independent Researcher

ORCID: 0009-0007-0190-9143

dejesuselias10@gmail.com

September 2026

Purpose

This addendum accompanies the note *The Sturmian–Mahler Edge of the Accelerated Collatz Realizer Problem* [3] (hereafter **the note**) without altering it. The note’s §9 audits the literature gate around the 2-adic value $\Xi_{\alpha,\beta}$ theorem by theorem, stating in each case the hypothesis that excludes the Collatz evaluation point; it omitted the closest prior work on the object itself, the two papers of López and Stoll. This addendum adds them, supplies the dictionary identifying $\Xi_{\alpha,\beta}$ with the value they study, and qualifies the one statement in §11 that the omission left overstated.

No theorem, proof or numerical result in the note changes, and this addendum makes no new mathematical claim; nothing here asserts or bears on irrationality of $\Xi_{\alpha,\beta}$, which remains the note’s Open Problem 1.

1 The omitted citations, and what they contain

López–Stoll 2009 [4]

The $3x+1$ conjugacy map over a Sturmian word, *Integers* **9** (2009), #A13, 141–162.

For every irrational slope $\alpha \in (0, 1)$ with convergents p_k/q_k and $1c_\alpha(j) = \lceil (j+1)\alpha \rceil - \lfloor j\alpha \rfloor$, their Theorem 1 gives, in $\mathbb{Z}_2$,

$$\Phi(1c_\alpha) = -\frac{1}{3} - \sum_{j \geq 0} (-1)^{j+1} \frac{2^{q_{j+1}+q_j-1}}{3(3^{p_{j+1}} - 2^{q_{j+1}})(3^{p_j} - 2^{q_j})}, \quad (1)$$

Φ being the Bernstein–Lagarias conjugacy map [2]; their Corollary 2 gives a generalized continued fraction for $-1/\Phi(1c_\alpha)$, convergent in $\mathbb{Z}_2$. Their §4 then leaves $\mathbb{Z}_2$ and studies the real function $\Phi_{\mathbb{R}}$, proving **irrationality of the real values** $\Phi_{\mathbb{R}}(m_\alpha)$ for $\alpha \in \mathbb{Q} \cap (\ln 2 / \ln 3, 1]$, and of a dual limit point for $0 < \alpha < \ln 2 / \ln 3$. The domain restriction is forced: $\sum 2^{nq}/3^{np}$ converges in $\mathbb{R}$ if and only if $\ln 2 / \ln 3 < p/q \leq 1$.

On the 2-adic question they claim nothing. Their abstract says the examples “*suggest* that Φ always maps Sturmian words to infinite words of full complexity”; their §4 says of the transcendence analogue, “*We have no proof.*”

López–Stoll 2021 [5]

The $3x+1$ periodicity conjecture in $\mathbb{R}$, arXiv:2101.12747.

Their Theorem 1: if the trajectory of some $\zeta \in \mathbb{Q}_{\text{odd}}$ is divergent then $\lim h/\ell = \ln 2 / \ln 3$ exactly — the critical density is the only place a rational divergent trajectory could live — while their aperiodicity theorem requires $\lim(h/\ell) > \ln 2 / \ln 3$ **strictly**. At $\alpha = \ln 2 / \ln 3$ the real series diverges: their §7 Lemma 27 names it “the divergent series $-\Phi_{\mathbb{R}}(1c_\alpha)$ ”, and their §10 Lemma 41 computes its terms $t_i = 2^{\lfloor (i-1)/\alpha \rfloor} / 3^i \in (1/6, 1/3]$, which under $i = j+1$ are literally the summands of the note’s $\Xi_{\alpha,0}$ up to sign.

2 Dictionary

Signs were checked term by term against the note’s own Theorem 4.1, and the identity was verified numerically modulo 2^{3000} .

	the note	López–Stoll
slope	$\alpha = \log_2 3 \approx 1.58496$	$\alpha_{\text{LS}} = \ln 2 / \ln 3 \approx 0.630930$ (ones-density)
relation	$\alpha_{\text{LS}} = 1/\alpha$	
word	$S_i = \lfloor i\alpha + \beta \rfloor$; here $\beta = 0$	$1c_{\alpha_{\text{LS}}}(j) = \lceil (j+1)\alpha_{\text{LS}} \rceil - \lceil j\alpha_{\text{LS}} \rceil$
constant	$\Xi_{\alpha,0} = -\sum_{i \geq 0} 3^{-(i+1)} 2^{\lfloor i\alpha \rfloor}$	$\Phi(1c_{\alpha_{\text{LS}}})$
identity	$\Xi_{\alpha,0} = +\Phi(1c_{\ln 2 / \ln 3})$	

Sign check. Remark 4.2 of the note evaluates $(1, 2)^\infty$ to -5 ; the convention above reproduces that value, and the sign is opposite to the companion carry-constant note, where $\Xi_\alpha = -\Phi(1c_{\ln 2 / \ln 3})$. In parity-vector coordinates, therefore, $\Xi_{\alpha,\beta}$ is a value of the Bernstein–Lagarias conjugacy map [1, 2] on a Sturmian parity word, and at $\beta = 0$ it is exactly the value whose 2-adic expansion (1) describes. This identification belongs in §4, immediately after Theorem 4.1.

3 Addition to the literature gate (§9)

The following belongs as a further bullet in §9, in the same form as the others — the hypothesis that excludes our point, not an impression.

López–Stoll 2009 [4]; López–Stoll 2021 [5]. The closest prior work on this exact object. The 2009 paper gives the 2-adic series and continued-fraction expansion of $\Phi(1c_\alpha)$ for every irrational slope — the approximants and approximation depths of this edge — and then proves irrationality of the real values $\Phi_{\mathbb{R}}(m_\alpha)$ on $\mathbb{Q} \cap (\ln 2 / \ln 3, 1]$, with a dual construction below $\ln 2 / \ln 3$. The 2021 paper proves aperiodicity for ones-density *strictly* above $\ln 2 / \ln 3$, and shows that the critical density is the only possible home of a rational divergent trajectory. *Excluded by the domain hypothesis, at both places.* The real series converges exactly when $\ln 2 / \ln 3 < p/q \leq 1$, and the staircase F diverges at $x = \ln 2 / \ln 3$; our slope is that single excluded point, and it is excluded from the dual construction as well. On the 2-adic question the 2009 paper claims nothing. The exclusion mechanism is the resonance of Theorem 6.5(ii), reached independently by the authors of the closest prior work: at the critical slope there is no archimedean value to talk about.

This strengthens §9’s conclusion rather than weakening it: the note argues that the failure of the archimedean theory at $\Xi_{\alpha,\beta}$ is relocation, not degeneracy (Remark 6.6), and López and Stoll reach the same boundary from the opposite direction, recording the divergence explicitly.

4 Qualified claim

Original wording (§11, Remark 11.1).

“Irrationality does not follow from Sturmian aperiodicity: ‘rational 2-adic $\Rightarrow$ eventually periodic valuation word’ is the open periodicity conjecture [3], not a theorem.”

Corrected wording.

“Irrationality does not follow from Sturmian aperiodicity: ‘rational 2-adic $\Rightarrow$ eventually periodic valuation word’ is the periodicity conjecture [3], which is open at our slope. It is not open everywhere: López and Stoll [5] prove it for parity words of ones-density *strictly* greater than $\ln 2 / \ln 3$, and their Theorem 1 shows that a rational divergent trajectory would have density exactly $\ln 2 / \ln 3$. The critical density — the density of the mechanical words of slope $\alpha = \log_2 3$ studied here — is precisely the case their method does not reach.”

The remainder of Remark 11.1 — the equivalence between rationality of $\Xi_{\alpha,\beta}$ and a mechanical integer orbit in a generalized $3x + q$ system, and the statement that Corollary 5.6 settles only $q = 1$ — is unchanged and unaffected.

Open Problem 1. The statement stands verbatim. It gains one sentence of context: the archimedean analogue is a theorem off the critical slope — López and Stoll [4] prove irrationality of the real values $\Phi_{\mathbb{R}}(m_\alpha)$ for $\alpha \in \mathbb{Q} \cap (\ln 2 / \ln 3, 1]$, and of the dual limit point below $\ln 2 / \ln 3$ — and neither covers $\alpha = \ln 2 / \ln 3$, where the real series diverges. The 2-adic problem is therefore not a corollary of their results.

Bibliographic note. The note’s bibliographic note (p. 10) lists which references were verified against primary or publisher sources. The entries added here — [4], [5], [2], [1] — were read in full from the primary sources: the 2009 paper (22 pp., *Integers* **9**, #A13) and the 2021 preprint (51 pp., arXiv:2101.12747).

Acknowledgment. The omission was identified during an adversarial citation audit of the Sturmian thread, assisted by Claude (Anthropic). All statements here, and any errors, are the author’s.

References

- [1] Daniel J. Bernstein. A noniterative 2-adic statement of the $3N+1$ conjecture. *Proceedings of the American Mathematical Society*, 121(2):405–408, 1994.
- [2] Daniel J. Bernstein and Jeffrey C. Lagarias. The $3x+1$ Conjugacy Map. *Canadian Journal of Mathematics*, 48(6):1154–1169, 1996.
- [3] Elias De Jesús. The Sturmian–Mahler Edge of the Accelerated Collatz Realizer Problem, 2026. technical note, June 2026.
- [4] Josefina López and Peter Stoll. The $3x+1$ Conjugacy Map over a Sturmian Word. *Integers*, 9(A13):141–162, 2009.
- [5] Josefina López and Peter Stoll. The $3x+1$ Periodicity Conjecture in $\mathbb{R}$, 2021.